Advanced Statistics 2027-1

Probabilistic Machine Learning

Course number: 367-2-5561

Lecturer	Opher Donchin
Tutorials	None
Communication	By e-mail
Office hours	By appointment

Syllabus

This course builds **working competence with probabilistic machine-learning methods**. Students learn to recognize a meaningful learning problem, formulate it probabilistically, select and use an appropriate computational approach, evaluate its predictions or inferences, and explain what the method does and does not justify. Mathematical derivations provide the shared reasoning that makes a repertoire of algorithms transferable rather than a catalogue of recipes. The goal is not abstract coherence alone, and not operating algorithms without understanding their assumptions.

The primary reference is Murphy's *Probabilistic Machine Learning: Advanced Topics* (**Book 2**); Book 1 supplies selected background and scaffolding. The course progression builds a toolkit for probabilistic modeling and evaluating performance; parameter fitting from complete and incomplete data; posterior approximation and exact structured inference; two lectures on learned neural representations for images and sequences; and latent generation. L07 and L09 compare different approaches to inference on the same Challenger thermal-distress posterior.

Each meeting includes approximately two hours of mathematical lecture and one hour of guided notebook work. Lectures develop the ideas, derivations, use conditions and comparisons. Notebooks supply **bounded practical use** on recognizable datasets: run or complete a method, inspect outputs, compare an informative baseline, and diagnose task-relevant errors. Small exact examples remain valuable as diagnostics but need not substitute for substantive machine-learning experience.

Software progression. Students enter with experience in Python, NumPy, PyMC, Bambi and ArviZ. The course uses NumPy/SciPy for probability, numerical checks and introductory inference, and introduces PyTorch in L05 for automatic differentiation and subsequent neural or variational fitting. JAX is not required.

Additional materials

1. Kevin P. Murphy, *Probabilistic Machine Learning: Advanced Topics*, MIT Press, 2023 (Book 2). Online edition.
2. Kevin P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022 (Book 1). Online edition.
3. Student-facing lecture materials, Jupyter notebooks and course resources: Advanced Statistics course repository.

Prerequisites

There are no formal prerequisites. The expected preparation is a mathematical probability course, basic linear algebra, Python programming and a Bayesian-first undergraduate statistics course (such as 367-1-4361 Statistical Computation and Analysis). Students should have encountered basic regression, DAGs, hierarchical Bayesian models, and posterior inference/diagnostics with PyMC, Bambi and ArviZ; the course extends rather than reteaches them. Students without equivalent preparation should consult the instructor.

Course requirements and grading

The regular course grade is based on three components totaling 100%:

Component	Contribution
InClass notebook submissions	15%
Project	30%
Final exam	55%

InClass notebooks. Submit the completed Jupyter notebook (`.ipynb`) through Moodle by **Wednesday at 23:59 (Israel time)** for each lecture. Grading of these submissions will be automated and will include AI-driven assessment of students' answers. Students may use AI freely in preparing their work, provided they paste their answers into the submitted notebook themselves (honor system). Grades will be returned through Moodle.

AI tutoring extra credit. For each lecture, submitting the AI-based tutoring activity by the next lecture earns an additional 0.2 percentage points toward the final course grade. A later submission before the final exam earns 0.1 percentage points. The two bonuses are alternatives for the same lecture, not cumulative.

Schedule

Planned 13-meeting sequence. Topics, notebook activities and readings may be adjusted during the semester.

Lecture 1

21.10.2026

Probabilistic ML and Gaussian systems

Topics Machine-learning task framing; probabilistic models; Bayes review; covariance and correlation; multivariate Gaussians; conditioning; missing-data inference; linear Gaussian systems.

Book 2
- 2.1.5–2.1.6 Conditional probability; Bayes
- 2.2.1.1 Bernoulli and binomial
- 2.2.4.1 Beta distribution
- 2.3.1.1 Gaussian joint distribution
- 2.3.1.3–2.3.1.5 Gaussian conditionals; information form
- 2.3.2.1–2.3.2.2 Linear-Gaussian systems
- 3.2.1 Coin-flipping Bayes
- 3.4–3.4.1 Conjugate priors; binomial

Book 1
- 1.1–1.5 ML tasks and datasets
- 2.3 Bayes' rule
- 2.4.1 Bernoulli; binomial
- 2.7.4 Beta distribution
- 3.1.1–3.1.3 Covariance; correlation
- 3.2.1–3.3.4 Multivariate and linear Gaussians
- 4.3–4.3.1 Empirical risk
- 4.6.1–4.6.2 Beta-binomial Bayes

Notebook Gaussian inference to impute missing measurements.

Skills Multivariate NumPy linear algebra and cross-checking posterior computations.

Lecture 2

28.10.2026

Exponential families and learning from sufficient statistics

Topics Exponential-family representation; Gaussian canonical form; derivatives of the log-partition function; natural and mean parameters; MLE as moment matching; maximum entropy; Gamma fit; categorical counts; Markov-chain learning.

Book 2
- 2.2.1.2 Categorical distribution
- 2.4.1–2.4.2 Exponential-family form; examples
- 2.4.3–2.4.4 Log partition; moments
- 2.4.5 MLE; moment matching
- 2.4.7 Maximum entropy
- 2.6.1–2.6.3 Markov chains; language models

Book 1

Notebook Estimate a character-bigram model and generate names.

Skills Probability estimation and numerical scoring with NumPy/SciPy.

Lecture 3

04.11.2026

Graphical models: factorization and conditional independence

Topics Derive DAG factorization from the probability chain rule and independence assumptions; work the student network and parameter economy. Develop d-separation and directed Markov blankets. Define undirected potentials and global normalization, compare separation and independence expressivity, and interpret the binary Ising model. Image posterior and algorithm derivations belong to the notebook.

Book 2
- 4.2.1–4.2.2 DAG factorization; student network
- 4.2.4 / selected 4.2.8 Independence; plates
- 4.3.1–4.3.2.1 / 4.3.6.1 Undirected models; Ising; separation
- 4.5.1 Directed/undirected independence comparison
- 4.3.8 Image restoration: notebook

Book 1
- Optional 3.6.1–3.6.2 Graphical-model orientation

Notebook Construct an Ising prior and Gaussian observation likelihood; derive the joint, posterior and single-site conditional. Implement or complete sequential local maximization (ICM) in a NumPy scaffold; compare with a pixelwise baseline and examine initialization or local-optimum limitations.

Skills Build a probabilistic restoration model from local factors, derive its posterior update, and distinguish local maximization from a global optimum or posterior marginal inference.

Lecture 4

11.11.2026

Information theory: mutual information, processing and noisy coding

Topics
Finite entropy, cross-entropy and KL divergence; derive mutual information and sketch the data-processing inequality with its conditional-independence premise. State source-length bounds. Work three-bit parity detection and posterior decoding, distinguishing information, Bayes message risk and a coding guarantee.

Book 2
- 5.1.3 / 5.1.6; 5.2.1 / 5.2.4 Information and uncertainty
- 5.3.1–5.3.4 Shared information and processing
- 5.4.1 Code-length bounds: statement only
- 5.5 Noisy-channel example: three-bit parity model

Book 1
- Selected 6.1 / 6.2.1–6.2.2 / 6.3 Uncertainty and shared information

Notebook
Evaluate prepared Iris probability forecasts using held-out mean log score and accuracy, against a supplied class-frequency baseline. Interpret supplied reliability and bounded information/representation examples. No classifier training or population mutual-information estimation is required. The live three-bit parity example provides an exactly checkable information/error contrast.

Skills
Choose and interpret probabilistic performance measures on real predictions, distinguishing scoring, prediction error and information retained.

Lecture 5

18.11.2026

Generalized linear models, automatic differentiation and SGD

Topics
Start from the familiar Bayesian regression likelihood and parameter prior; explain why Gaussian observations are unsuitable for binary responses and counts, without implying that the hierarchical model fails. Generalize the observation family using Gaussian/Bernoulli/Poisson GLMs and links. Contrast full posterior inference with MLE/MAP point fitting; derive the canonical NLL and vector softmax gradients, then reverse-mode autodiff and minibatch SGD.

Book 2
- 15.1 / 15.1.1 / 15.1.3 Common model structure and derivative
- 15.1.2 Brief alternative-link comparison
- 15.3.2 Three-class probabilities
- Selected 6.2; 6.3.1–6.3.2 Automatic differentiation and batch updates

Book 1
- Selected 7.8.2 / 7.8.5 Matrix derivatives
- Selected 8.2 / 8.4.1 / 8.4.3 Gradient and batch-update support

Notebook
Fit the three-class Iris softmax model in PyTorch; compare the manual mean-batch gradient with autograd at identical parameters, run visible SGD updates and evaluate held-out log loss/probabilities. Data preparation and program scaffolding are supplied.

Skills
Connect a likelihood derivative with reverse-mode automatic differentiation and a checked stochastic parameter-update/evaluation loop.

Lecture 6

25.11.2026

Latent-variable learning and expectation-maximization

Topics
Recall Bayesian latent-variable hierarchies and joint posterior inference, then distinguish point estimation by marginal likelihood from inference over all unknowns. Show why unknown latent values complicate direct fitting, and derive the ELBO/KL identity, exact E-step tightness and EM monotonicity. Apply Gaussian conditional first/second moments to unknown mean/covariance with missing data; briefly interpret mixture responsibilities and assumptions.

Book 2
- Selected 6.5.1 / 6.5.3 Probability bound and EM
- 6.5.4 Gaussian missing-data updates
- 3.11 When the missing-data mechanism can be ignored
- 28.2.1 Group-assignment probabilities: brief example

Book 1
- Selected 3.5 / 8.7 Grouped observations and EM explanation

Notebook
Assemble expected Gaussian sufficient statistics, including conditional covariance, and perform an E/M update inside a supplied iteration driver. Check observed likelihood and missing-entry reconstruction; no second full mixture fit.

Skills
Use conditional uncertainty in sufficient-statistic estimation and distinguish an exact EM improvement guarantee from arbitrary approximate hidden-value updates.

Lecture 7

02.12.2026

Inference overview and Laplace approximation

Topics
Posterior queries, local Gaussian approximation, checked numerical reference and sample-average error. Derive scalar Laplace and iid estimator variance; interpret dependent-sampler diagnostics under their conditions.

Book 2
- 7.1–7.3 Inference questions and method choice
- Selected 7.4.1–7.4.3 Mode, numerical reference and Laplace
- 11.2–11.2.2 Sample averages and iid error; selected 12.6.3 diagnostics: interpretation

Book 1

Notebook
Use the 23-flight Challenger thermal-distress logistic posterior, with standardized temperature and proper Gaussian prior. Compare point and Laplace predictions with a supplied posterior sampler and checked two-dimensional reference. Work posterior-predictive averages and query-specific numerical error. Fitting, reference and diagnostics helpers supplied; no sampler construction or claim of exact MCMC output.

Skills
Choose and check posterior answers; distinguish proxy error, posterior spread and finite sampling error.

Lecture 8

09.12.2026

Exact inference in structured sequence models

Topics
Scaled categorical-HMM filtering, smoothing and evidence; joint path MAP versus marginal decisions; dense $O(T\,K^2)$ message cost. Compact Gaussian filtering comparison.

Book 2
- 9.2.1–9.2.6 HMM messages, scaling and path queries
- Selected 8.1.2 / 8.2.2 Gaussian filtering

Book 1

Notebook
Explore exact conditional inference for Gaussian time series using a Kalman-filter notebook. Work with conditional means and covariances, model fitting and predictive checks using provided code.

Skills
Transfer exact conditional-inference logic across categorical and Gaussian sequences.

Lecture 9

16.12.2026

Variational and amortized inference

Topics
ELBO/reverse KL, Gaussian family restriction, total pathwise gradients and optimizer error. Introduce the shared evidence-to-variational-parameters map and recognize coordinate-ascent updates.

Book 2
- 10.1.1–10.1.2 Variational criterion
- 6.3.3 / 6.3.5; 10.2.1.1 Pathwise gradients
- 10.3.1 Coordinate rule; 10.1.5 amortization

Book 1

Notebook
Fit a Gaussian variational approximation to the Challenger thermal-distress posterior; compare specified predictive queries with Laplace, supplied sampling output and a checked reference. Use a known correlated Gaussian to separate family and optimization error.

Skills
Fit and diagnose a variational posterior and explain amortized repeated-query computation.

Lecture 10

23.12.2026

Deep Neural Networks I: learned representations and images

Topics
GLM to learned nonlinear composition; categorical probability output; shared local filters, equivariance and receptive fields; one residual-gradient argument. Connect these architectures to automatic differentiation and SGD.

Book 2
- 16.1 Two probabilistic roles
- 16.2.1–16.2.5 Linear, nonlinear, convolutional, residual and normalization blocks
- 16.3.1–16.3.2 MLP and CNN composition

Book 1
- 13.1–13.2.5 Learned features and depth
- Selected 13.3–13.5 Training support; 14.2.1 convolution

Notebook
Source-adapted PyTorch LeNet on Fashion-MNIST: connect a local filter calculation to package output, inspect shapes/sharing, run bounded updates and evaluate held-out log score and errors against a supplied dense baseline. Prepared data, training code and checkpoint are provided; no architecture search.

Skills
Explain, fit and evaluate a learned compositional image representation.

Lecture 11

30.12.2026

Deep Neural Networks II: representing sequence context

Topics Autoregressive likelihood, shared recurrent state and bounded unrolled derivative; Jacobian products; worked causal attention and positional information; context/cost tradeoffs and model versus inference-network roles.

Book 2
- 16.2.7–16.2.8 Attention and recurrence
- 16.3.4–16.3.5 RNN and transformer composition

Book 1
- 15.2.1 / 15.2.4–15.2.6 RNN and training
- Selected 15.4–15.5 Attention, positions and transformers

Notebook Source-adapted PyTorch character RNN on Time Machine text. Compare held-out token log score with a smoothed Markov baseline, trace hidden-state updates and resets, then contrast teacher-forced scoring with categorical generation. Supply corpus splits, driver and checkpoint; no second transformer training.

Skills Model sequence context and choose a computational architecture under explicit dependency/cost assumptions.

Lecture 12

06.01.2027

Variational autoencoders: amortized inference and rate–distortion

Topics Generative model versus amortized posterior; joint encoder/decoder objective with event-summed Gaussian NLL and diagonal KL, then observation averaging. Reuse ELBO/pathwise/neural results. Derive rate as mutual information plus aggregate-posterior mismatch; interpret beta, collapse and nonidentifiability.

Book 2
- Selected 21.1–21.2 Model, inference network and training objective
- 21.3.1 Changing the penalty weight
- Selected 5.4.2 Reconstruction/information tradeoff
- Selected 21.4 When the model ignores hidden coordinates

Book 1

Notebook Fit or complete a small mobility-measurement VAE, compare prior generation with reconstruction, inspect latent descriptions and evaluate two controlled reconstruction/rate tradeoffs.

Skills Use and critique a latent generative model for reconstruction, generation and descriptive representations without overinterpreting hidden factors.

Lecture 13

13.01.2027

Project presentations and course synthesis

Topics Student project presentations connecting probabilistic assumptions, computation, prediction, evaluation, uncertainty and scientific interpretation.

Book 2

Book 1

Notebook Student project presentations.

Skills Scientific presentation and computational reproducibility.